# MARIANA MACEDO SANTOS

Data Science • Data Engineering

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## Summary

Computer Science student (UFOP, graduating 12/2026) with hands-on experience in data engineering: I build ETL pipelines in Python and SQL, orchestrated with Apache Airflow, at TerraLab. I also have machine learning and NLP projects and did an exchange in Industrial Data Science in Austria.

## Education

**B.Sc. in Computer Science** **2022 – 12/2026**
Federal University of Ouro Preto (UFOP), Brazil

**Exchange Student – Industrial Data Science** **02/2026 – 07/2026**
Montanuniversität Leoben, Austria

- Hands-on specialization in Industrial Data Science, applied Machine Learning and Process Optimization.

**IT Technician (technical high school degree)** **2019 – 2021**
Federal Institute of Northern Minas Gerais (IFNMG), Brazil

## Experience

**Data Analytics – TerraLab, UFOP** **08/2026 – Present**

- Develop and optimize ETL pipelines in Python and SQL to load and process large volumes of data.
- Manage and monitor environments with Apache Airflow (orchestration), Apache Doris (data warehouse) and Apache Superset (dashboards).

**Undergraduate Research Assistant – Data Automation and Pipelines** **UFOP, 2023**

- Developed Python scripts to automate data extraction, cleaning and integration, streamlining operational routines.

**Vice-President – Computer Science Student Association** **UFOP, 2024 – 2026**

- Organized academic events and technical workshops, managed student teams and represented students.

## Projects

**Brazilian Economic Indicators Pipeline** *Airflow, PostgreSQL, SQL, Superset, Docker*

- Daily Airflow pipeline that extracts five indicators from the Central Bank of Brazil API, organizes the data in layers (raw, staging and analytics) in a PostgreSQL data warehouse and feeds a Superset dashboard, with incremental, idempotent loads.

**Undergraduate Thesis – Automatic Classification of Programming Error Messages** *Python, scikit-learn*

- Classification model (TF-IDF and Logistic Regression) that maps C compilation errors to programming topics, as a basis for an educational recommender system.

**Transphobic Hate Speech Detection with BERTimbau** *Python, PyTorch, Transformers*

- Collected tweets through the X API and fine-tuned a Portuguese BERT model to classify them, as a pair project.

**Video Transcription App** *Node.js, Express, PostgreSQL, Firebase, Next.js*

- REST API in Node.js/Express with a layered architecture, Firebase authentication, PostgreSQL persistence, asynchronous file processing and per-user usage quotas.

## Technical Skills

- **Data & Pipelines:** SQL, PostgreSQL, Relational Modeling, ETL, Apache Airflow, Apache Doris.
- **Analysis & Visualization:** Pandas, NumPy, R, statistical analysis, Power BI, Apache Superset, Matplotlib.
- **Machine Learning & NLP:** scikit-learn, PyTorch, Hugging Face Transformers (BERT), TF-IDF.
- **Software Engineering:** Git/GitHub, Docker, REST APIs, Python (Flask), Node.js (Express).
- **Programming Languages:** Python, SQL, R, JavaScript/TypeScript, Java, C/C++.
- **Courses:** Pandas for Data Analysis (Kaggle), Advanced Python (Udemy), REST APIs with Flask and Python (Udemy).
- **Languages:** Portuguese (native), English (advanced).